Nonlinear Modeling of Cycling

By Christina Cavalluzzo

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Background

Longest and middle leg of a triathlon

- Half Ironman: bike 56 miles

Goal for the bike

- Maximize speed
- Preserve legs for the run



Mechanical Power Model

$$F_{total} = F_{roll} + F_{slope} + F_{accel} + F_{aero}$$

$$F_{total} = mgC_{rr} + mgsin(\theta) + ma + \frac{1}{2}\rho C_d A V^2$$

$$P_{total} = F_{total} * V$$

$$P_{total} = mgC_{rr}V + mgsin(\theta)V + maV + \frac{1}{2}\rho C_d A V^3$$ [1, 3]

Muscle Power Model

Hill equation: $(v + b)(F + a) = b(F_0 + a)$ [2]

$$P_{muscle} = F(v) * v$$

$$F(v) = \frac{b(F_0 + a)}{v + b} - a$$

$$P_{muscle}(v) = \left(\frac{b(F_0 + a)}{v + b} - a\right) * v$$

Equations Coupled

Power balance:

$$\left(\frac{b(F_0 + a)}{v + b} - a\right) * v = mgC_{rr}V + mgsin(\theta)V + maV + \frac{1}{2}\rho C_d A V^3$$

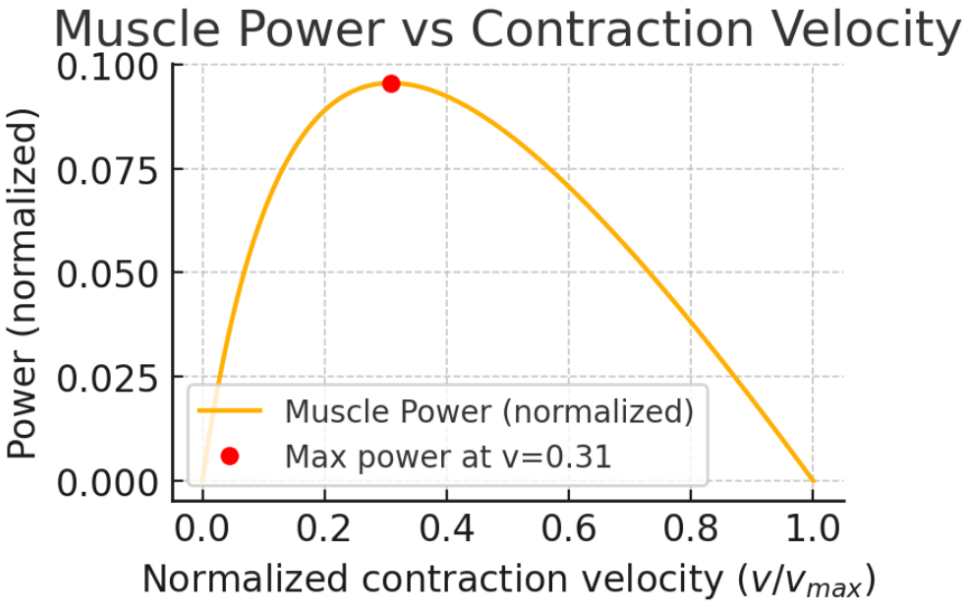
$$m\dot{V} = \frac{1}{r}\left(\frac{b(F_0 + a)}{v + b} - a\right) - \left[mgC_{rr} + mgsin(\theta) + ma + \frac{1}{2}\rho C_d A V^2\right]$$ [3]

Equations and Coupling

$$\dot{x} = V$$

$$\dot{V} = \frac{1}{r}\left(\frac{b(F_0 + a)}{v + b} - a\right) - \left[mgC_{rr} + mgsin(\theta) + ma + \frac{1}{2}\rho C_d A V^2\right] \quad [3]$$

$$v = \frac{V}{r} \quad [4]$$



[2]



Assumptions and Parameters

Cycling Positions



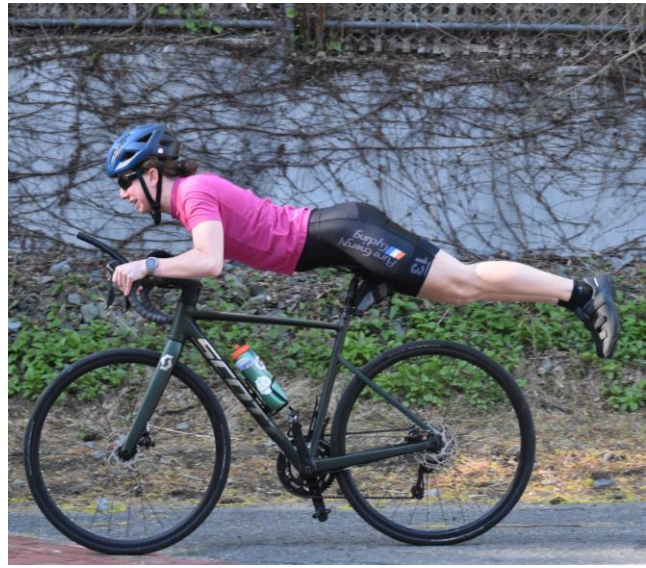
Hands On Hoods



Hands In Drop Bars



Hands In Aero Bars



Plank



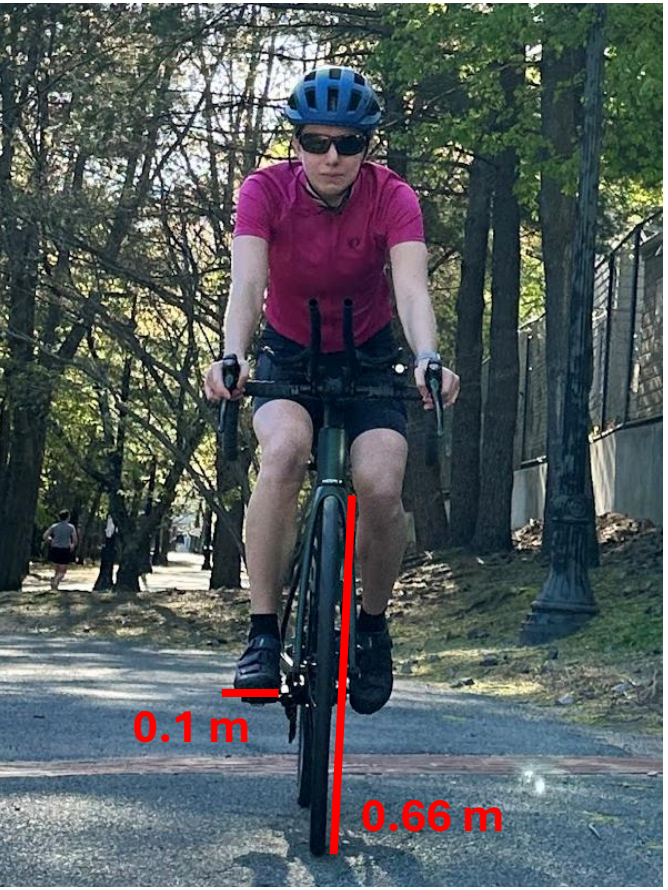






Area Measurements

— 0.1 m



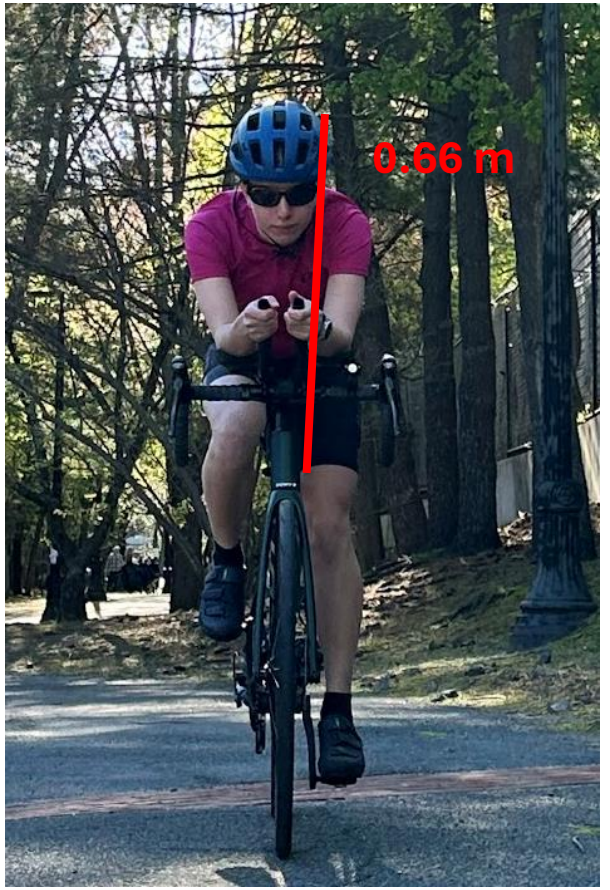
Hands On Hoods

Area ~ 0.363 m^2



Hands In Drop Bars

Area ~ 0.343 m^2



Hands In Aero Bars

Area ~ 0.323 m^2



Plank

Area ~ 0.118 m^2

Strava



Constants

Weight (cyclist + bike) = 69.4 kg

Coefficient of rolling resistance, Cr = 0.003

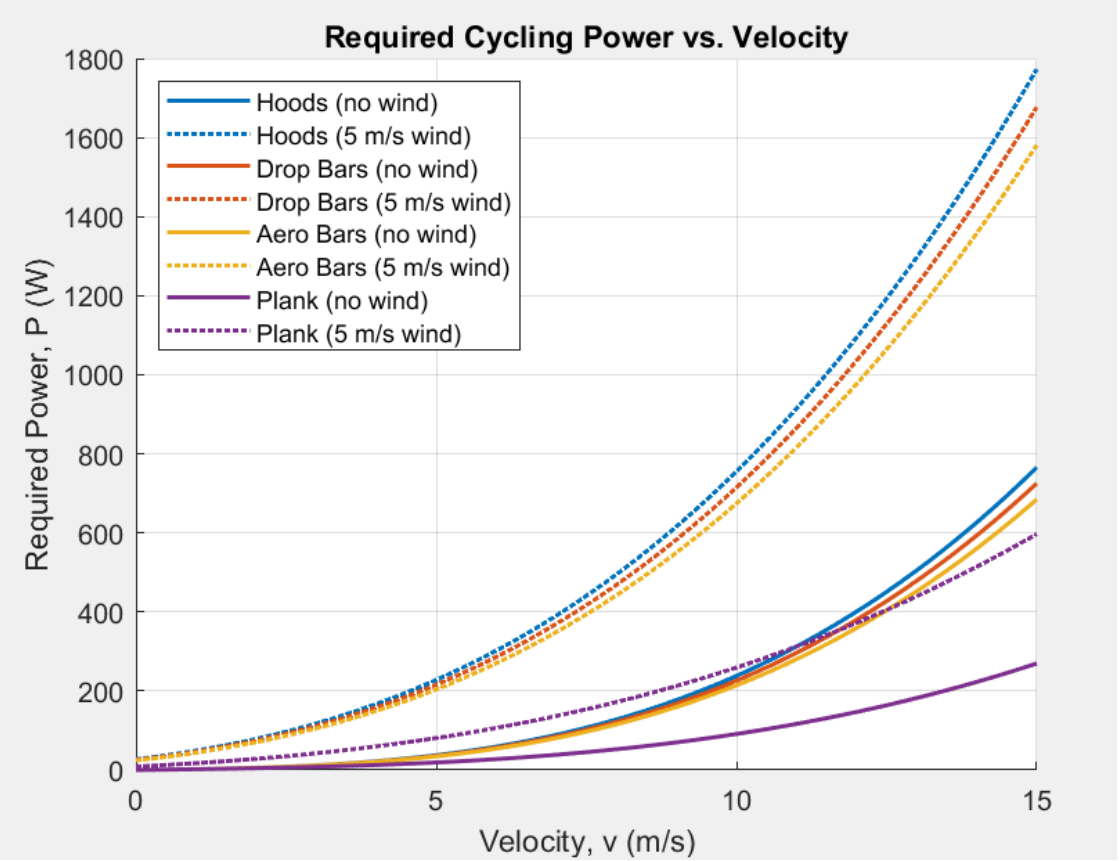
Density of air, r = 1.2 kg/m^3

Drag coefficient, Cd = 0.9

Terrain slopes: 0, +5 (uphill), -5 (downhill) degrees

Muscle power output, Pmuscle = 250W

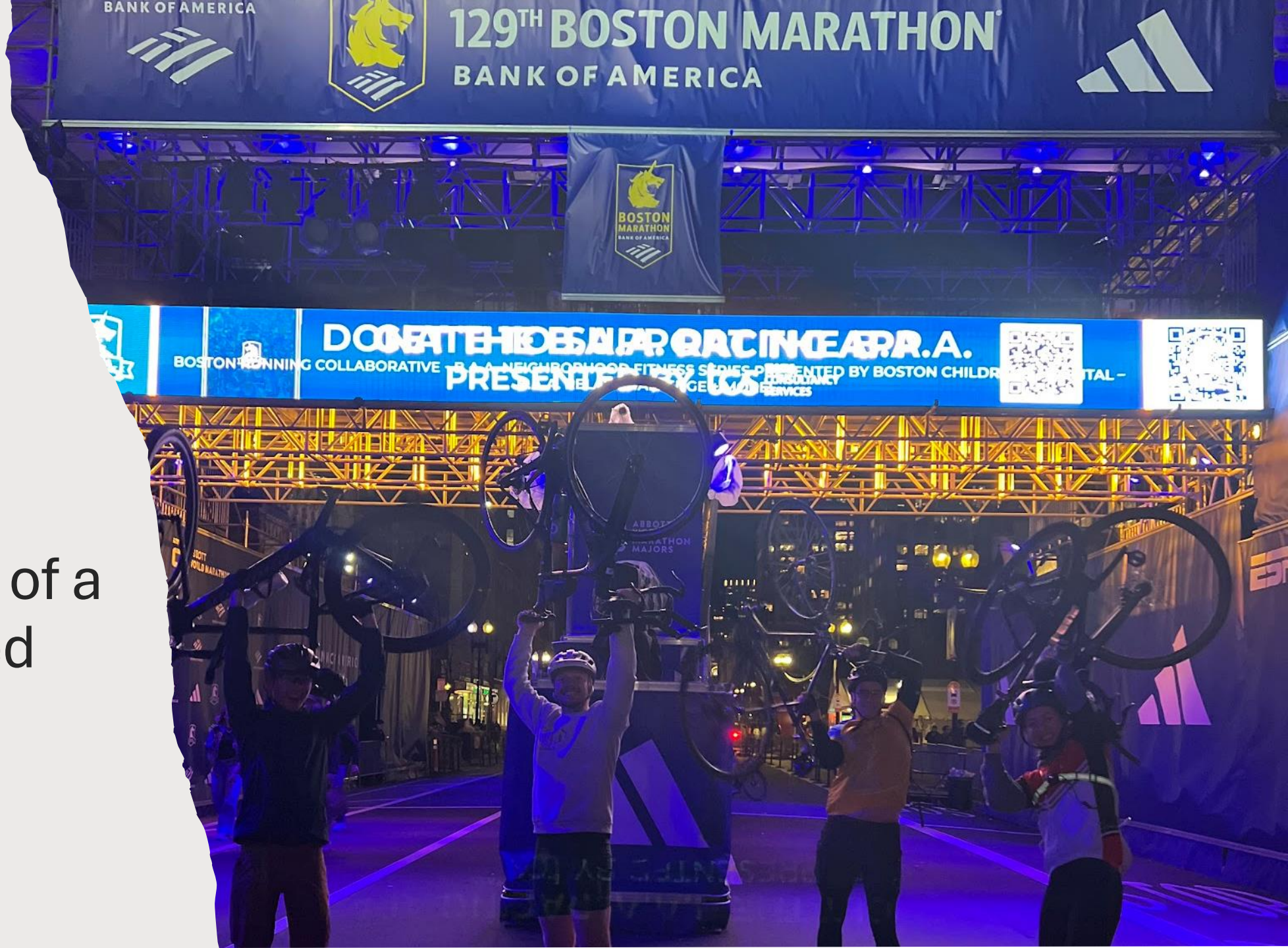
*Optimal gear assumption: rider is always in a gear that produces optimal cadence



Steady State Velocity (Equilibrium Points)

	Hoods (A=0.363)	Drop Bars (A=0.343)	Aero Bars (A=0.323)	Plank (A=0.118)
Flat (0°)	23.49 mph	23.94 mph	24.38 mph	33.78 mph
Uphill (+5°)	8.72 mph	8.75 mph	8.83 mph	-
Downhill (-5°)	42.50 mph	43.62 mph	44.74 mph	71.58 mph

CL Stability of a PI-Regulated Model



State Space Model

State Eqns

$$\dot{x}_1 = \frac{P_{mus}(t) - P_{tot}(x_1 + V_{ref}, i)}{m(x_1 + V_{ref})} \longrightarrow \dot{x}_1 = \frac{u - P_{tot}(V)}{mv}$$

$$\dot{x}_2 = -x_1$$

$$P_{tot}(V) = \alpha V + \beta V^3$$

Linearization

$\dot{x_1} = \frac{u - P_{tot}(V)}{mV}$

u is normalized muscle effort

$x_1 = V - V_{ref}$ (speed deviation)

$\frac{\partial f}{\partial V} = -\frac{\alpha + 3\beta V_{ref}^2}{mV_{ref}}, \qquad \frac{\partial f}{\partial u} = -\frac{1}{mV_{ref}}$

$a = -\frac{\alpha + 3\beta V_{ref}^2}{mV_{ref}} < 0, \mathrm{b} = \frac{1}{mV_{ref}} > 0$

$\dot{x_2} = -x_1$ is already linear

So,

$$\dot{x_1} = ax_1 + b\tilde{u}$$
$$\dot{x_2} = -x_1$$

PI Control

PI Controller

$$u(t) = u^* - K_p x_1 + K_i x_2$$

Small deviation from equilibrium

$$\tilde{u} = u - u^* = -K_p x_1 + K_i x_2$$

CL linear dynamics

$$\dot{x_1} = (a - bK_p)x_1 + bK_i x_2$$
$$\dot{x_2} = -x_1$$

Routh-Hurwitz

$$A_{CL} = \begin{pmatrix} a - bK_p & bK_i \\ -1 & 0 \end{pmatrix}$$

Characteristic eqn:

$$\lambda^2 - (a - bK_p)\lambda + bK_i = 0$$

Routh-Hurwitz:

$$s^2 + a_1 s + a_0$$

$$a_1 = -(a - bK_p) \qquad a_0 = bK_i$$

Conclusion: continuous PI loop around posture's equilibrium is a.s. for $a_1 > 0, a_0 > 0$

Bifurcations



Varying Slope

Equilibrium:

$$P_{mus} = mgC_{rr}V + mgsin(\theta)V + \frac{1}{2}\rho C_d A V^3$$

$$P_{tot} = \gamma_1(\theta)V + \gamma_2 V^3$$

Where $\gamma_1 = mgC_{rr} + mgsin(\theta)$, $\gamma_2 = \frac{1}{2}\rho C_d A$

$$p_{mus}^{max} = \gamma_1 V(\theta) + \gamma_2 V^3$$

At the bifurcation:

$$\frac{dP_{tot}}{dV} = 0 \quad \text{and} \quad P_{tot}(V) = p_{mus}^{max}$$

Solve to get

$$\gamma_1 + 3\gamma_2 V^2 = 0 \longrightarrow \mathrm{V}^2 = \frac{-\gamma_1}{3\gamma_3}$$

To get real V, RHS >= 0. Bifurcation at $\gamma_1 = 0$:

$$\theta_c = -\arcsin(C_{rr})$$

Conclusion: saddle-node bifurcation

Varying Drag Area

Equilibrium:

$$P_{mus} = mgC_{rr}V + mgsin(\theta)V + \frac{1}{2}\rho C_d A V^3$$

$$P_{tot} = \gamma_1(\theta)V + \gamma_2 V^3$$

Where $\gamma_1 = mgC_{rr} + mgsin(\theta)$,

$$\gamma_2 = \frac{1}{2}\rho C_d A$$

$$p_{mus}^{max} = \gamma_1 V + \gamma_2 V^3$$

At the bifurcation:

$$\frac{dP_{tot}}{dV} = 0 \quad \text{and} \quad P_{tot}(V) = p_{mus}^{max}$$

Solve to get

$$\gamma_1 + 3\gamma_2 V^2 = 0 \longrightarrow V^2 = \frac{-\gamma_1}{3\gamma_3}$$

Critical A:

$$A_c = -\frac{2mg(C_{rr} + \sin(\theta))}{3\rho C_d V^2}$$

Conclusion: saddle-node bifurcation

Varying Integral Gain

Linearized System:

$$\dot{x_1} = (a - bK_p)x_1 + bK_i x_2$$

$$\dot{x_2} = -x_1$$

$$\rightarrow \quad \dot{x_1} = A_{CL}x$$

where

$$A_{CL} = \begin{pmatrix} a - bK_p & bK_i \\ -1 & 0 \end{pmatrix}$$

Eigenvalues

$$\lambda^2 - (a - bK_p)\lambda + bK_i = 0$$

Hopf:

Eigenvalues purely imaginary $\lambda = \pm jw$

Happens when $a - bK_p = 0$ and $bK_i > 0$

So, $K_p = \frac{a}{b}$ and $K_i > 0$

Critical gain

$$K_p = \frac{a}{b} \text{ where a } = \frac{\frac{dP_{tot}}{dV}}{mV_{ref}}$$

Sustained oscillations begin at

$$\lambda = \pm j\sqrt{bK_i}$$

Varying Force Limit, F_0

Max muscle power output:

$$P_{mus} \leq F_0 V$$

Compared to required power:

$$P_{tot} = \gamma_1 V + \gamma_2 V^3$$
$$F_0 V = \gamma_1 V + \gamma_2 V^3$$
$$F_0 = \gamma_1 + \gamma_2 V^2$$
$$V^2 = (F_0 - \gamma_1)/\gamma_2$$

If

$F_0 < \gamma_1$ then $V^2 < 0$ then no forward equilibrium

Conclusion: saddle-node bifurcation, critical bifurcation at $F_0^{crit} = \gamma_1 = mg(C_{rr} + \sin\theta)$

Implementing a Controller

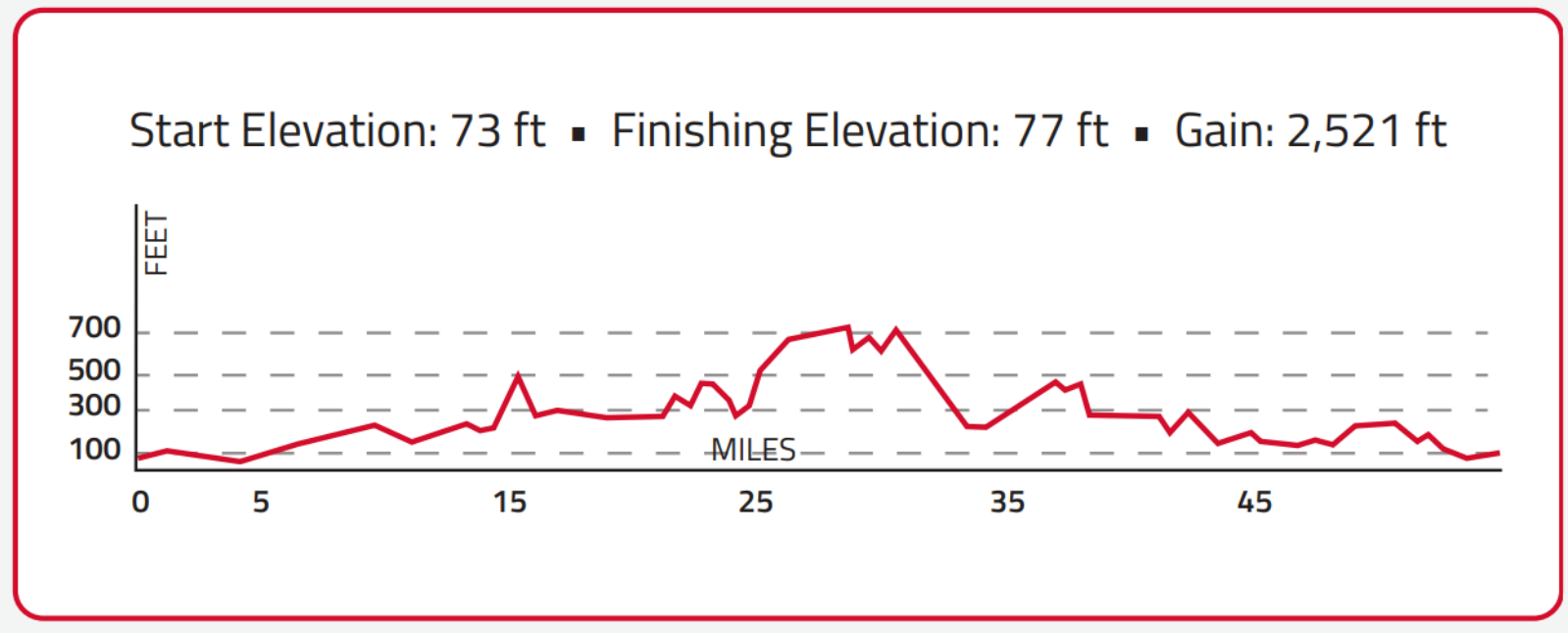


Adaptive Controller for Model Course

Parameters: 56 miles, start at rest and no stopping

Repeating Terrain: Synthetic course cycling through 0°, +5°, -5° every 5 miles

Western Mass Ironman 70.3: simulating the actual race course terrain



[5]

Controller 1 – steady pacing + moderate effort

$$\dot{V} = \frac{1}{mr} F_{muscle}(v)\alpha(t) - \frac{1}{m}\left[mgC_{rr} + mgsin(\theta) + \frac{1}{2}\rho C_d A V^2\right]$$

Constant effort pacing strategy

α = effort (0 to1)

=> strategy: initial α = 1 , then settles into steady effort α = 0.8

```
acc = (F_drive - F_resist) / m;
V = V + acc * dt;
if V < 0
    V = 0;
end
distance = distance + V * dt;
time = time + dt;

power = F_drive * V;
V_mph = V * 2.23694;
```

```
if strcmpi(controller_type, 'steady')
    alpha = 0.8;
    if posture == PLANK
        alpha = 0;
    end
```

Controller 2 – dynamic adaptive pacing

Change power based on slope and speed

Uphills: α = 1

Downhills: back off and coast (sometimes α = 0)

Threshold: if downhill slope is < 2°, light pedaling

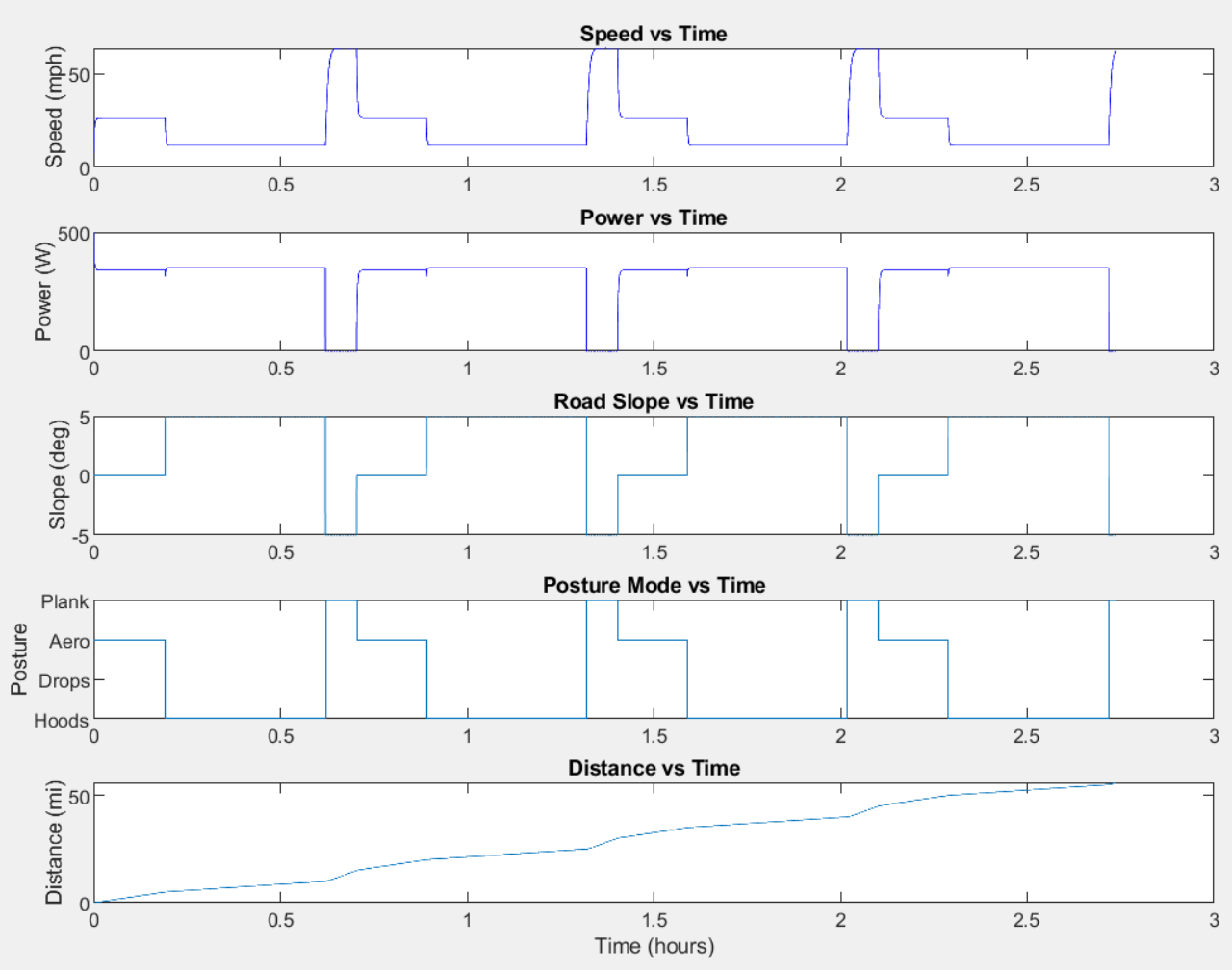
Flat: α ~ 0.85

```
if strcmpi(controller_type, 'steady')
    alpha = 0.8;
    if posture == PLANK
        alpha = 0;
    end
else
    if theta_deg >= 1
        alpha = 1.0;
    elseif theta_deg >= -1
        alpha = 0.8;
    elseif theta_deg >= -3
        alpha = 0.5;
    else
        alpha = 0.0;
    end
end
```

```
if theta_deg > 3
    posture = HOODS;
elseif theta_deg > 1
    posture = DROPS;
elseif theta_deg > -3
    posture = AERO;
else
    posture = PLANK;
end
```

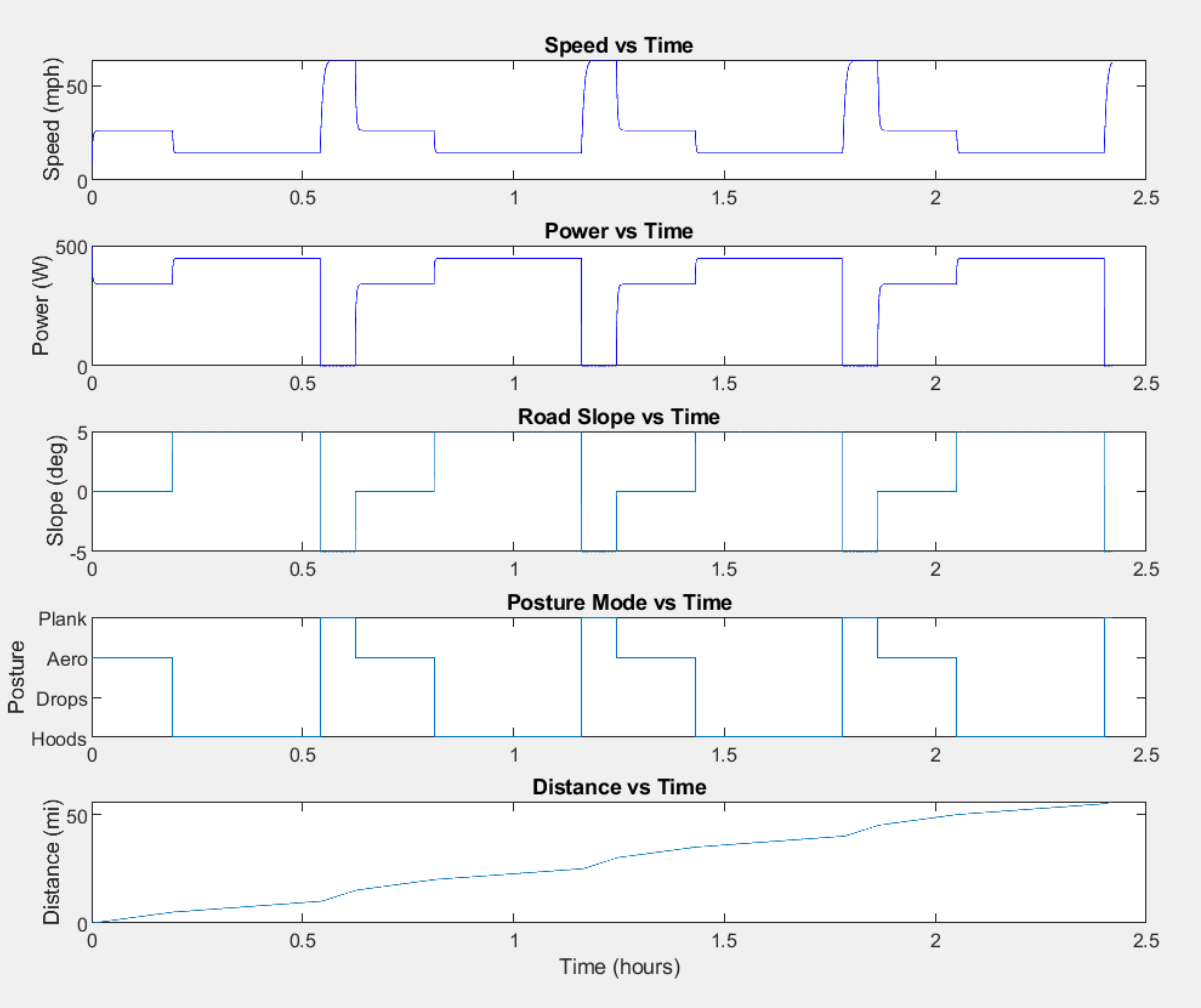
```
>> simulation_trial1
Controller: steady, Course: repeating
Total time to complete 56 miles: 2 h 44 min 16.0 s
Average speed: 20.45 mph (9.14 m/s)
Average power output (while pedaling): 345.5 W
```

Repeating hills, steady




```
>> simulation_trial1
Controller: adaptive, Course: repeating
Total time to complete 56 miles: 2 h 25 min 11.0 s
Average speed: 23.14 mph (10.34 m/s)
Average power output (while pedaling): 408.9 W
```

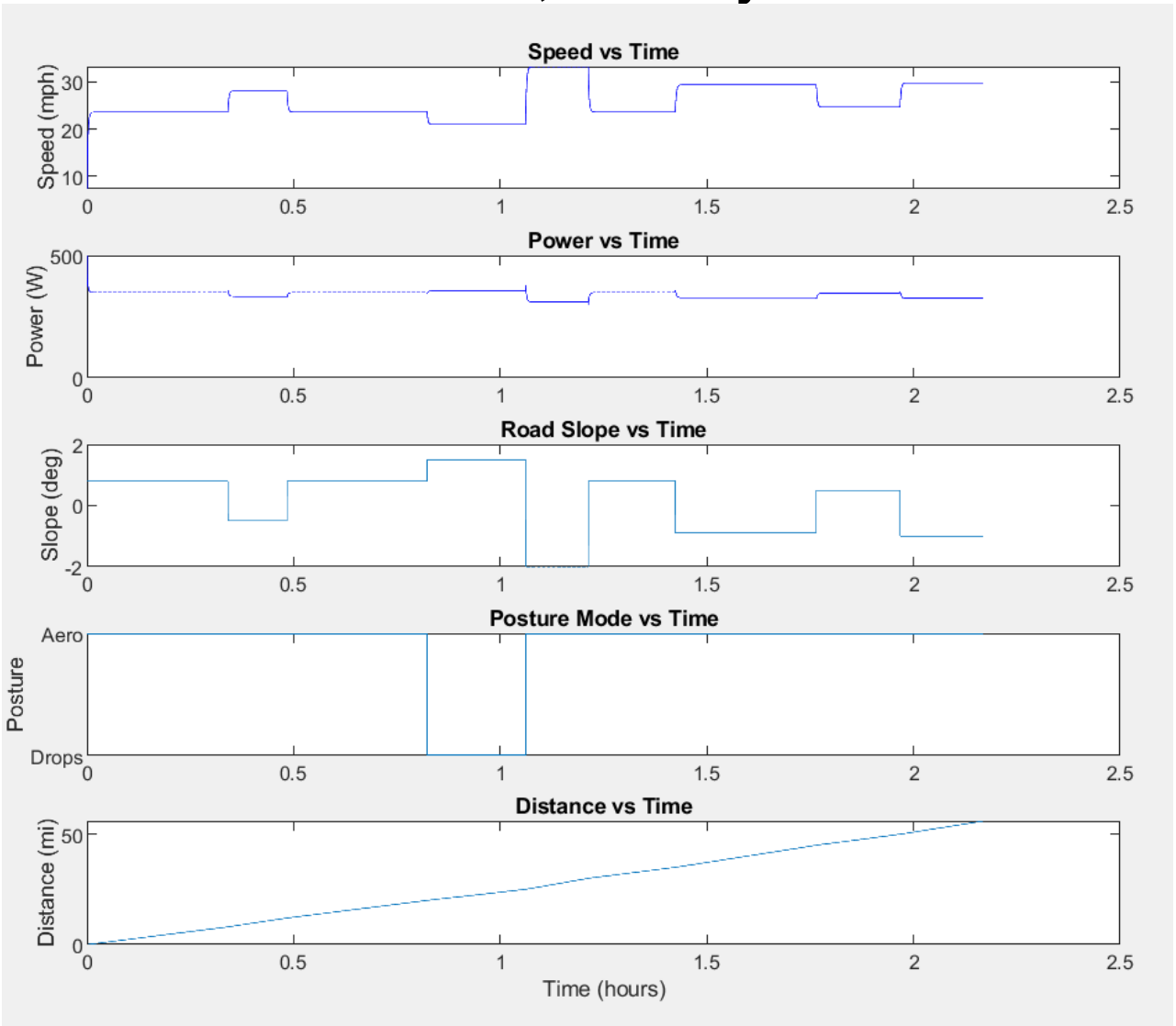
Repeating hills, adaptive



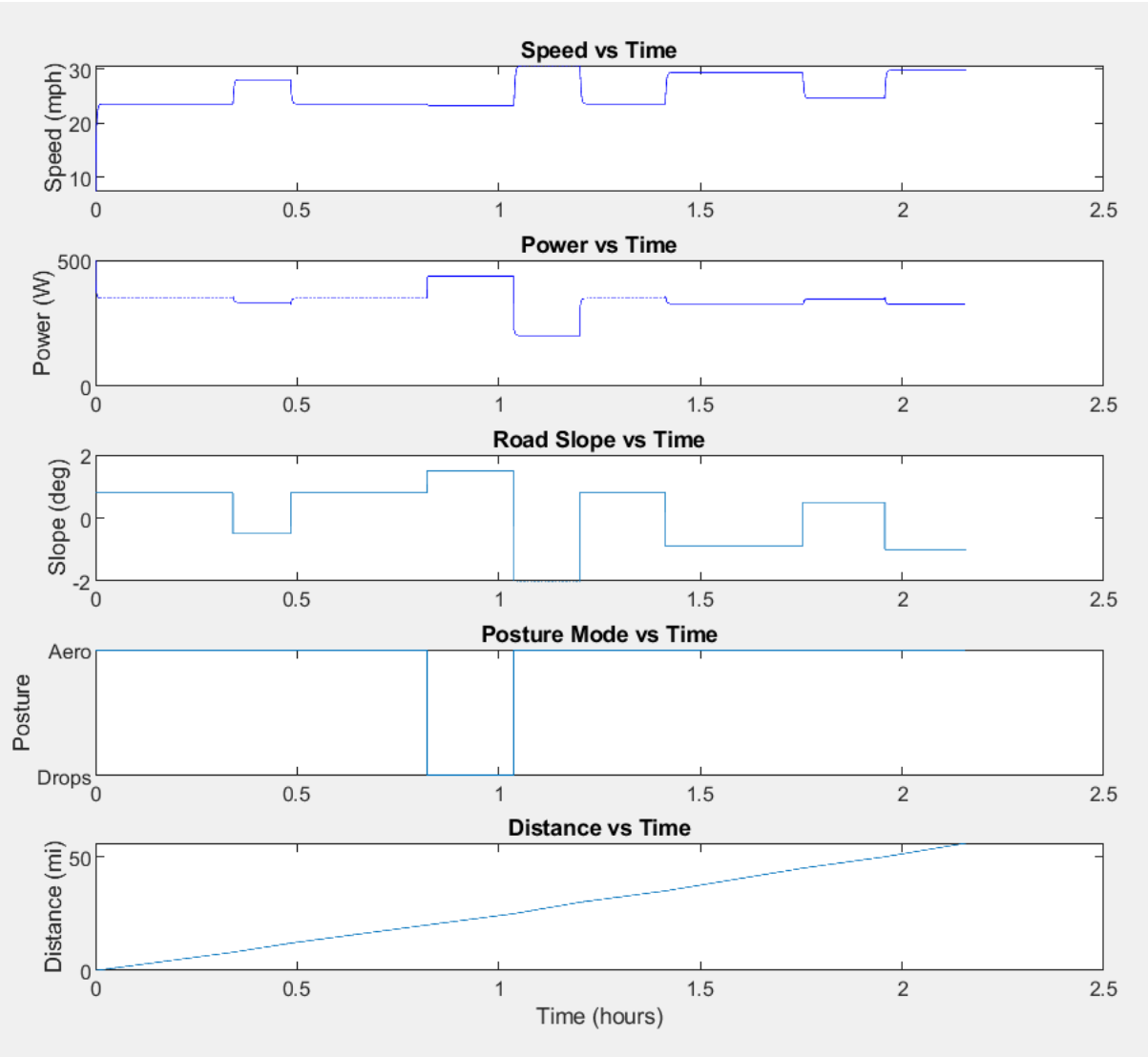
```
>> simulation_trial1
Controller: steady, Course: ironman
Total time to complete 56 miles: 2 h 10 min 9.0 s
Average speed: 25.81 mph (11.54 m/s)
Average power output (while pedaling): 340.1 W
```

```
>> simulation_trial1
Controller: adaptive, Course: ironman
Total time to complete 56 miles: 2 h 9 min 30.0 s
Average speed: 25.94 mph (11.60 m/s)
Average power output (while pedaling): 339.6 W
```

Ironman, steady



Ironman, adaptive

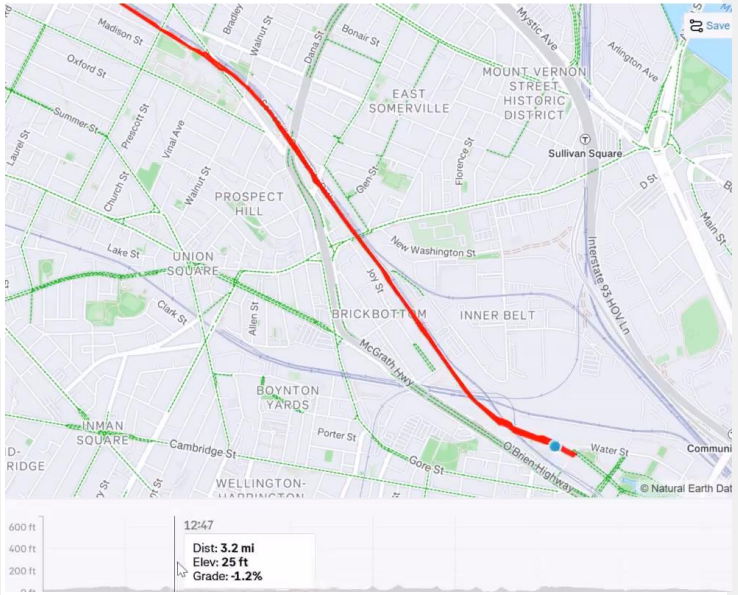


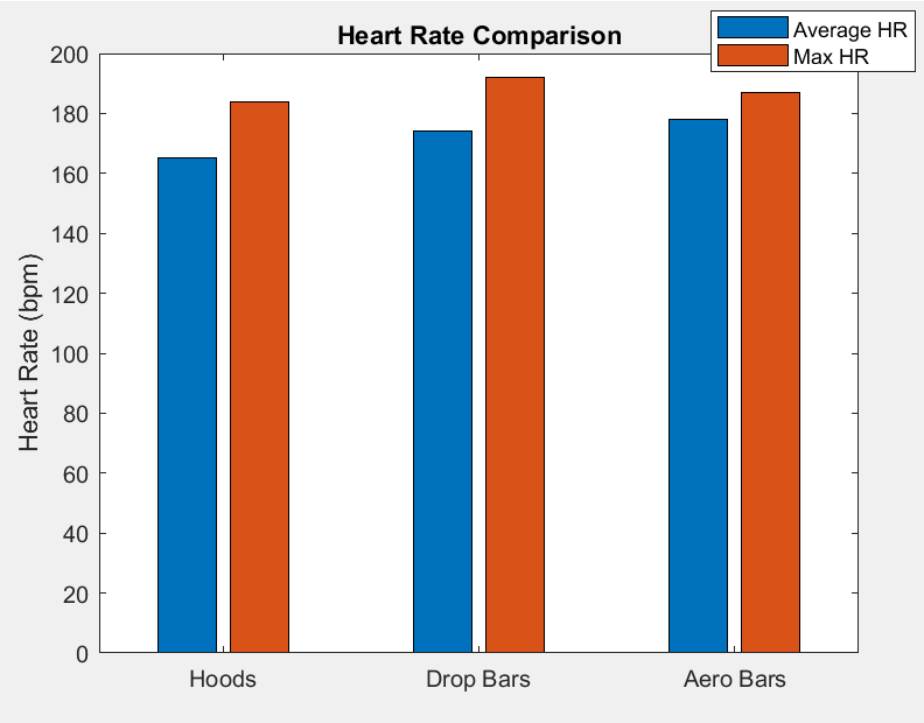
Practical Recommendations for Performance Enhancement

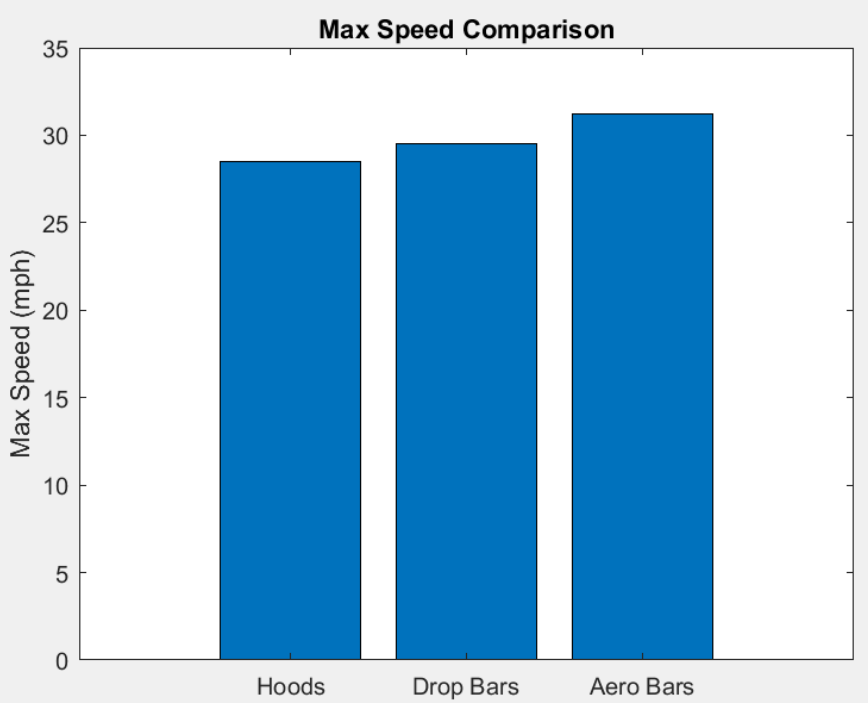
- Use aero bars whenever at highest speeds, and whenever possible
- For hilly terrain, adopt adaptive pacing (the cyclist is their own controller!)
- Average power output is best when modulated, not constant. Work the uphills, cruise on the downhills

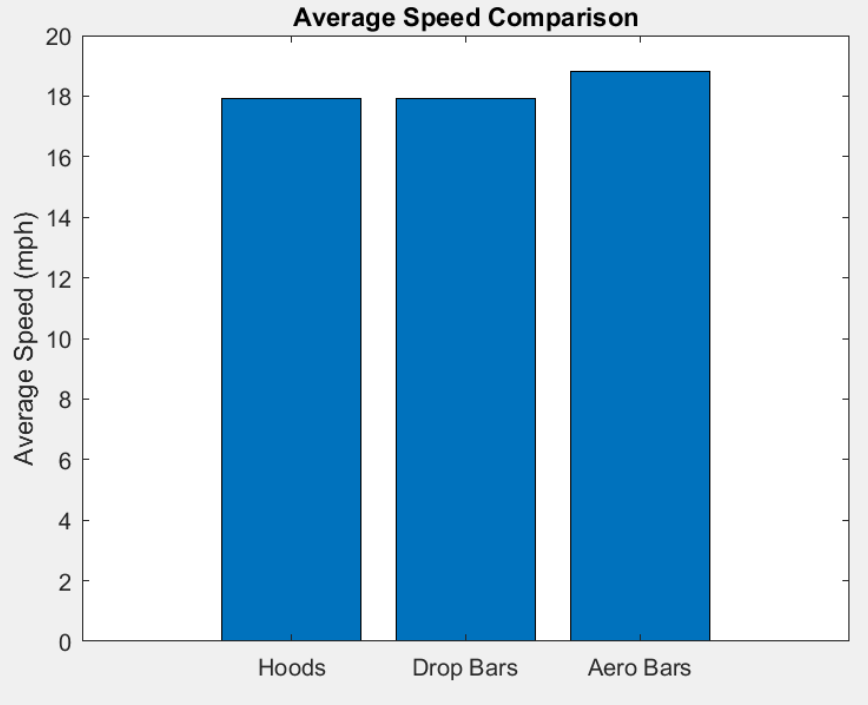
Experimentation

- Data collection
- Garmin vivoactive 4S









Future works

- Heart rate modeling
- VO2 max
- Applying to training
- Incorporate wind/ environmental conditions

References

[1] Rinard, D., & Allen, J. (n.d.). Bicycle Drag Force Formulas (R. Pivit, Ed.). Www.sheldonbrown.com; Klara Agil. https://www.sheldonbrown.com/rinard/aero/formulas.html

[2] Muscle Tension-Velocity. (n.d.). Www.brown.edu. https://www.brown.edu/Departments/Engineering/Courses/En123/Lectures/HillEqn.htm

[3] Muelaner, J. (1998, August 7). Experiments Prove Equation for Bicycle Power. Better Bicycles. https://betterbicycles.org/science/speed/validation-of-bicycle-power-equation/

[4] Dahmen, T. (n.d.). Computing a Field of Optimal Pacing Strategies for Cycling Time Trials Introduction, Related Work, and Target Application. https://www.uni-konstanz.de/mmsp/pubsys/publishedFiles/Da13a.pdf

[5] Course Overview. (2025). IRONMAN. https://www.ironman.com/races/im703-western-massachusetts/course


Questions?